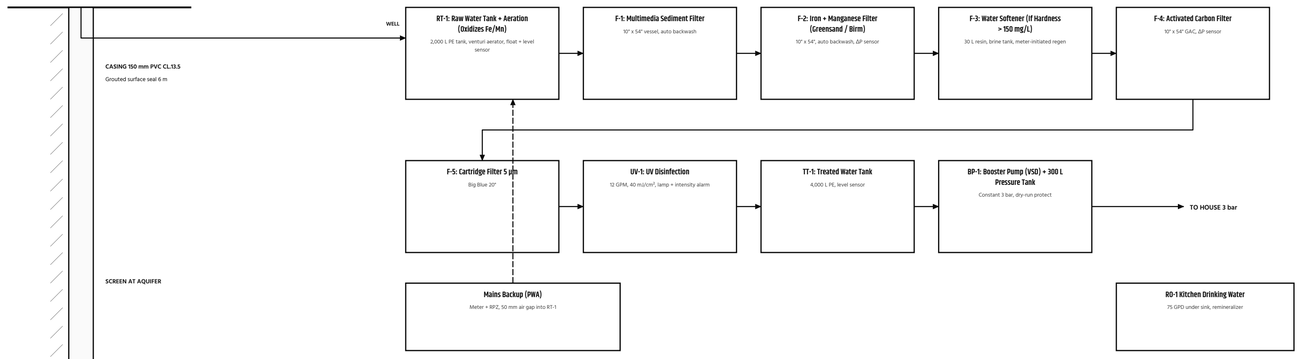
WATER SOURCE PRIORITY + TREATMENT SCHEMATIC

1. Rainwater
Garden cisterns T-1/T-2/T-3 for irrigation only

2. Drilled Well
Primary house supply, via treatment

3. Mains (PWA)
Backup into RT-1 via 50 mm air gap + RPZ



1 **TYPICAL DEEP WELL**
SCALE: NTS

SIZING + PRIORITY

Demand: 6 people x 200 L + pool/plunge make-up ≈ 1.5 m^3/day; peak 60 L/min
Well depth: 60-120 m (typical Thai deep well), confirm by DGR hydro-survey
Casing: 150 mm (6") PVC Class 13.5 + steel surface casing, grouted 6 m
Pump: Submersible 1.5 kW (2 HP), 3 m^3/h @ 60 m head, dry-run + low-level cut-out

Storage: 2,000 L raw (aeration, contact time) + 4,000 L treated (≈ 2.5 days) + 300 L pressure tank. Cisterns T-1,T-3 stay garden-only.

WATER TEST FIRST: Lab water test (Fe, Mn, hardness, TDS, pH, nitrate, arsenic, coliform) BEFORE final treatment selection; about 3,000-5,000 THB. Iron/manganese and softening stages are installed only if the test shows they are needed.

Controls: well pump on RT-1 level; ΔP alerts call a backwash on solar surplus; UV fault closes the house valve and alerts; tank levels on the dashboard (E-103).

ROUGH COST (THB)

ITEM	THB
Drilling 100 m incl. casing, gravel pack, grout, development	180,000
Submersible pump, controller, cable, wellhead	40,000
Treatment train (RT-1 to BP-1)	251,000
Kitchen RO	15,000
Mains backup, RPZ, air gaps, piping, install	70,000
Water test	5,000
Total	561,000

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE
Design Development

NOT FOR CONSTRUCTION

DATE
11 Oct 2026

SCALE
NTS

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

WATER NOTES

Well permit from the DGR (Groundwater Act) for wells > 15 m deep; licensed driller.

Backflow: RPZ on mains, air gaps on every top-up, no cross-connection with garden water.

Locate the well > 30 m from the drain field and septic.

SHEET TITLE
Water Source + Treatment

SHEET
P-103